

1680. Concatenation of Consecutive Binary Numbers

Difficulty: Medium

Link: <https://leetcode.com/problems/concatenation-of-consecutive-binary-numbers/?envType=daily-question&envId=2026-02-28>

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Problem:

Given an integer n , return the **decimal value** of the binary string formed by concatenating the binary representations of 1 to n in order, **modulo** $10^9 + 7$.

Example 1:

Input: $n = 1$
Output: 1
Explanation: "1" in binary corresponds to the decimal value 1.

Example 2:

Input: $n = 3$
Output: 27
Explanation: In binary, 1, 2, and 3 corresponds to "1", "10", and "11".
After concatenating them, we have "11011", which corresponds to the decimal value 27.

Example 3:

Input: $n = 12$

Output: 505379714

Explanation: The concatenation results in "1101110010111011110001001101010111100".

The decimal value of that is 118505380540.

After modulo $10^9 + 7$, the result is 505379714.

Constraints:

- $1 \leq n \leq 10^5$